

# A finite-timestep covariance term in multidimensional Itô-2

Eric,

I think there is one small but real finite- $\Delta t$  issue in the multidimensional argument. Nothing changes in 1D. It also disappears in the strict  $\Delta t \rightarrow 0$  limit. But for the actual one-kick-per-hydro-step algorithm, independent coordinate kicks match the marginal variances, not the full MC covariance.

## Exact spot in the paper

I mean Section 2.2 of [arXiv:2604.23041v2](https://arxiv.org/abs/2604.23041v2), printed page 7.

The 1D result is fine. Equations (28)–(29) give

$$u_i = \frac{h_i}{\Delta t} C_{-,i}, \quad \kappa_{ii} = \frac{h_i^2}{2\Delta t} (C_{+,i} - C_{-,i}^2). \quad (1)$$

The  $-C_{-,i}^2$  term is the subtraction of the squared mean.

The issue starts at equations (45)–(46), where the paper calls

$$\kappa_{ij} = \frac{\mathbb{E}[\Delta X_i \Delta X_j]}{2\Delta t} \quad (2)$$

the covariance matrix and then uses

$$\mathbb{E}[\Delta X_i \Delta X_j] = \mathbb{E}[\Delta X_i^2] \delta_{ij}. \quad (3)$$

The second equation is right for the *raw* second moment. For a finite-step covariance, the matching generalization of equation (29) is

$$\boxed{\kappa_{ij}^{(\text{step})} = \frac{h_i^2 C_{+,i} \delta_{ij} - h_i h_j C_{-,i} C_{-,j}}{2\Delta t}}. \quad (4)$$

So the sentence immediately after equation (46), saying that the matrix is diagonal and the dimensions are independent, is only true for the raw second moment, or in the strict infinitesimal-rate limit.

Equations (47)–(48) can stay. Equation (49) should be written

$$\boxed{\Sigma \Sigma^\top = 2\kappa}, \quad (5)$$

not as an element-by-element square root, unless  $\kappa$  is diagonal.

Equation (59) can also stay:

$$\mathbf{X}^{n+1} = \mathbf{X}^n + \mathbf{u} \delta t + \Sigma \mathbf{W}_{\delta t}. \quad (6)$$

It just needs the full factor  $\Sigma$ . Footnote 4 sets  $\delta t = \Delta t$ , which is exactly where the finite-step cross covariance survives.

## The issue

One MC step chooses exactly one face, or stays put. So for  $i \neq j$ ,

$$\mathbb{E}[\Delta X_i \Delta X_j] = 0. \quad (7)$$

But that is the *raw* cross moment, not the covariance:

$$\text{Cov}(\Delta X_i, \Delta X_j) = \mathbb{E}[\Delta X_i \Delta X_j] - \mathbb{E}[\Delta X_i] \mathbb{E}[\Delta X_j] \quad (8)$$

$$= -\mathbb{E}[\Delta X_i] \mathbb{E}[\Delta X_j]. \quad (9)$$

Therefore, whenever the mean transport has two nonzero components,

$$\boxed{\text{Cov}(\Delta X_i, \Delta X_j) \neq 0 \quad (i \neq j).} \quad (10)$$

The 3D discussion calls  $\mathbb{E}[\Delta X_i \Delta X_j] / (2\Delta t)$  the covariance and then sets the off-diagonal entries to zero because a single MC jump cannot move in two directions. The second statement is right for the raw moment. The missing term is the product of the means.

## Full finite-step result

For direction  $i$ , let

$$p_i^+ = P(\Delta \mathbf{X} = +h_i \mathbf{e}_i), \quad p_i^- = P(\Delta \mathbf{X} = -h_i \mathbf{e}_i), \quad (11)$$

$$C_{+,i} = p_i^+ + p_i^-, \quad C_{-,i} = p_i^+ - p_i^-. \quad (12)$$

The remaining probability is the no-jump outcome:

$$P(\Delta \mathbf{X} = 0) = 1 - \sum_i C_{+,i}. \quad (13)$$

First moment:

$$m_i \equiv \mathbb{E}[\Delta X_i] = h_i C_{-,i}. \quad (14)$$

Raw second moment:

$$R_{ij} \equiv \mathbb{E}[\Delta X_i \Delta X_j] = h_i^2 C_{+,i} \delta_{ij}. \quad (15)$$

Central second moment:

$$Q_{ij} \equiv \mathbb{E}[(\Delta X_i - m_i)(\Delta X_j - m_j)] \quad (16)$$

$$= R_{ij} - m_i m_j. \quad (17)$$

So

$$\boxed{Q_{ij} = h_i^2 C_{+,i} \delta_{ij} - h_i h_j C_{-,i} C_{-,j}.} \quad (18)$$

Equivalently,

$$\boxed{\mathbf{Q} = \text{diag}(h_i^2 C_{+,i}) - \mathbf{m} \mathbf{m}^\top.} \quad (19)$$

Diagonal:

$$Q_{ii} = h_i^2 (C_{+,i} - C_{-,i}^2), \quad (20)$$

which is exactly the current 1D formula.

Off diagonal:

$$\boxed{Q_{ij} = -h_i h_j C_{-,i} C_{-,j}, \quad i \neq j.} \quad (21)$$

## Why the infinitesimal argument looks diagonal

For finite velocities,

$$m_i = u_i \Delta t = O(\Delta t). \quad (22)$$

Therefore,

$$m_i m_j = O(\Delta t^2), \quad (23)$$

and

$$\frac{m_i m_j}{\Delta t} = O(\Delta t) \rightarrow 0. \quad (24)$$

So the off-diagonal Kramers–Moyal *rate* vanishes as  $\Delta t \rightarrow 0$ . No problem there.

The finite-step algorithm is different: it takes one increment with  $\delta t = \Delta t$ , and the diagonal coefficient already keeps the finite-step correction  $-C_{-,i}^2$ . The cross terms  $-C_{-,i}C_{-,j}$  are the matching finite-step corrections off the diagonal.

## Simple 2D example

Uniform positive flow, equal cell widths, and

$$C_x = C_y = C. \quad (25)$$

The MC outcomes are

$$\Delta \mathbf{X} = \begin{cases} (h, 0), & P = C, \\ (0, h), & P = C, \\ (0, 0), & P = 1 - 2C. \end{cases} \quad (26)$$

Then

$$\mathbb{E}[\Delta X] = \mathbb{E}[\Delta Y] = hC, \quad (27)$$

$$\text{Var}(\Delta X) = \text{Var}(\Delta Y) = h^2 C(1 - C), \quad (28)$$

$$\text{Cov}(\Delta X, \Delta Y) = -h^2 C^2, \quad (29)$$

$$\text{Corr}(\Delta X, \Delta Y) = -\frac{C}{1 - C}. \quad (30)$$

At  $C = 0.1$ ,

$$\text{Corr}_{\text{MC}} = -0.111111. \quad (31)$$

AthenaK, 131,072 Itô-2 particles:

|                                   | measured                | finite-step MC          |
|-----------------------------------|-------------------------|-------------------------|
| $\text{Var}(\Delta X)$            | $8.7997 \times 10^{-5}$ | $8.7891 \times 10^{-5}$ |
| $\text{Var}(\Delta Y)$            | $8.7941 \times 10^{-5}$ | $8.7891 \times 10^{-5}$ |
| $\text{Cov}(\Delta X, \Delta Y)$  | $-2.45 \times 10^{-8}$  | $-9.77 \times 10^{-6}$  |
| $\text{Corr}(\Delta X, \Delta Y)$ | $-2.78 \times 10^{-4}$  | $-0.111111$             |

Exactly what independent coordinate kicks should do: right marginal variances, zero cross covariance.

## Proposed fix

If the claim is only “same mean and variance in each coordinate,” the current method is fine. I would just say that explicitly.

If the claim is “same first two finite-step displacement moments,” use the full  $\mathbf{Q}$ .

Factor

$$\mathbf{Q} = \mathbf{L}\mathbf{L}^\top, \quad (32)$$

draw

$$\mathbb{E}[\boldsymbol{\xi}] = 0, \quad \mathbb{E}[\boldsymbol{\xi}\boldsymbol{\xi}^\top] = \mathbf{I}, \quad (33)$$

and update

$$\boxed{\Delta\mathbf{X} = \mathbf{m} + \mathbf{L}\boldsymbol{\xi}.} \quad (34)$$

With the current notation,

$$\mathbf{u} = \frac{\mathbf{m}}{\Delta t}, \quad \boldsymbol{\kappa} = \frac{\mathbf{Q}}{2\Delta t}, \quad (35)$$

so the same update is

$$\Delta\mathbf{X} = \mathbf{u}\Delta t + \sqrt{2\Delta t} \boldsymbol{\kappa}^{1/2} \boldsymbol{\xi}. \quad (36)$$

Bounded, independent, zero-mean, unit-variance components of  $\boldsymbol{\xi}$  still work. The factor  $\mathbf{L}$  supplies the correlation. This matches the MC mean and the complete finite-step covariance; higher moments remain a separate question.

Bottom line: independent coordinate kicks get each coordinate’s mean and variance right, but not the full finite-step covariance. To match that too, include  $-h_i h_j C_{-,i} C_{-,j}$  and draw the kick through  $\mathbf{Q} = \mathbf{L}\mathbf{L}^\top$ .

*Am I reading that right?*

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